UNIVERSIDADE ESTADUAL DE MONTES CLAROS – UNIMONTES

CENTRO DE CIÊNCIAS EXATAS E TECNOLÓGICAS – CCET

DEPARTAMENTO DE CIÊNCIAS DA COMPUTAÇÃO – DCC

ATIVIDADE XXII

RAMON LOPES DE QUEIROZ

MONTES CLAROS – MG

JUNHO/2026

RAMON LOPES DE QUEIROZ

ATIVIDADE XXII

Atividade avaliativa apresentada para atendimento de requisito parcial para aprovação na disciplina Cálculo Integral e Diferencial do Curso de Graduação em Bacharelado em Sistemas de Informação – 2º período

Professor: Daniel Oliveira

MONTES CLAROS – MG

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Exercício 22-

1- $\int \frac{x^3+2}{(x-1)^2}\,dx = \int \frac{x^3+2}{x^2-2x+1}\,dx = \int (x+2)\,dx + \int \frac{3x}{(x-1)^2}\,dx$

$= \frac{x^2}{2} + 2x + \int \frac{3x}{(x-1)^2}$ $u = x-1 \to x = u+1$, $du = dx$

$(x^3+0x^2+0x+2) \mid x^2-2x+1$

$-x^3+2x^2-x \qquad x+2$

$2x^2-x+2$

$-2x^2+4x-2$

$+3x$

$= x+2+\frac{3x}{(x-1)^2}$

$= \frac{x^2}{2} + 2x + 3\cdot\int \frac{u+1}{u^2}$

$= \frac{x^2}{2} + 2x + 3\cdot\int \left(\frac{u}{u^2} + \frac{1}{u^2}\right)$

$= \frac{x^2}{2} + 2x + 3\cdot\int \left(\frac{1}{u} + u^{-2}\right)$

$= \frac{x^2}{2} + 2x + 3\cdot\left(\ln u + \left(\frac{u^{-1}}{-1}\right)\right)$

$= \frac{x^2}{2} + 2x + 3\ln|x-1| - \frac{3}{x-1} + C$

2-a $\int \frac{2x-3}{(x-1)^3}\,dx = \int \frac{2(u+1)-3}{u^3} = \int \frac{2(u+1)}{u^3} - 3\int \frac{1}{u^3}\,du$

$u = x-1 \to x = u+1$
$du = dx$

$= 2\int \left(\frac{u}{u^3} + \frac{1}{u^3}\right) - 3\int u^{-3}\,du$

$= 2\int \left(u^{-2} + u^{-3}\right) - 3\cdot\frac{u^{-2}}{-2} + K$

$= 2\cdot\left(\frac{u^{-1}}{-1} + \frac{u^{-2}}{-2}\right) - 3\cdot\frac{-2}{u^2} + K$

$= 2\cdot\left(\frac{-1}{x-1} - \frac{2}{(x-1)^2}\right) + \frac{6}{(x-1)^2} + K$

$= \frac{-2}{x-1} - \frac{4}{(x-1)^2} + \frac{6}{(x-1)^2} + K \Rightarrow \frac{-2}{x-1} + \frac{2}{(x-1)^2} + K$

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b- $\int \frac{x^4+x+1}{x^3-x}\,dx = \int x + \frac{x^2+x+1}{x^3-x}\,dx = \int x + \int \frac{x^2+x+1}{x^3-x}$

$$\begin{array}{l|l} x^4+0x^3+0x^2+x+1 & x^3-x \\ -x^4 \quad -x^2 & x \\ \hline x^2+x+1 & \end{array}$$

$= \frac{x^2}{2} + \int \frac{x^2+x+1}{x(x-1)(x+1)}$

$= \frac{x^2}{2} + \int \frac{A}{x} + \frac{B}{(x-1)} + \frac{C}{(x+1)}$

$= \frac{x^2}{2} + \int A(x-1)(x+1) + B(x)(x+1) + C(x)(x-1)$

$x^2+x+1 = A(x-1)(x+1) + Bx(x+1) + Cx(x-1)$

$= \frac{x^2}{2} + \int \frac{-1}{x} + \frac{3/2}{x-1} + \frac{1/2}{x+1}\,dx$

$x=0$

$0^2+0+1 = A(0-1)(0+1) + B0 + C0$

$= \frac{x^2}{2} - \ln x + \frac{3}{2}\ln|x-1| + \frac{1\ln|x+1|}{2} + C$

$1 = A(-1)(1)$

$A = -1$

$x=1$

$1^2+1+1 = A(0) + B(1)(1+1) + C(0)$

$B = \frac{3}{2}$

$x=-1$

$(-1)^2+(-1)+1 = A(0) + B(0) + C(-1)(-2)$

$C = \frac{1}{2}$

c- $\int \frac{x+1}{x(x-2)(x+3)}\,dx = \int \frac{A}{x} + \frac{B}{x-2} + \frac{C}{x+3}\,dx = \int A(x-2)(x+3) + B(x)(x+3) + C(x)(x-2)$

$= \int \frac{-1/6}{x} + \frac{3/10}{(x-2)} + \frac{-2/15}{(x+3)}$

$x=0$	$x=2$	$x=-3$	
$0+1 = A(0-2)(0+3) + B(0) + C(0)$	$2+1 = A(0) + B(2)(2+3) + C(0)$	$-3+1 = A(0) + B(0) + C(-3)(-3-2)$	$= -\frac{1}{6}\ln|x| + \frac{3}{10}\ln|x-2| -$
$1 = A(-6)$	$3 = B(10)$	$-2 = C(15)$	
$A = -\frac{1}{6}$	$B = \frac{3}{10}$	$C = \frac{-2}{15}$	$\frac{2}{15}\ln|x+3| + K$

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d. $\int \frac{2}{(x+2)(x-1)^2}dx = \int \frac{A}{x+2} + \frac{B}{(x-1)} + \frac{C}{(x-1)^2} = \int A(x-1)^2 + B(x+2)(x-1) + C(x+2)$

$x=-2$

$2 = A(-2-1)^2 + B(0) + C(0)$

$2 = 9A$

$A = \frac{2}{9}$

$x=1$

$2 = A(0) + B(0) + C(1+2)$

$C = \frac{2}{3}$

$= \int \frac{2/9}{(x+2)} + \frac{-2/9}{(x-1)} + \frac{2/3}{(x-1)^2}$

$\frac{2}{9}\ln|x+2| - \frac{2}{9}\ln|x-1| - \frac{2}{3(x-1)} + K$

$x=0$

$2 = A(-1)^2 + B(2)(-1) + C(2)$

$2 = \frac{2}{9} - 2B + 2 \cdot \frac{2}{3}$

$B = -\frac{2}{9}$

e. $\int \frac{x+3}{x^3-2x^2-x+2}dx = \int \frac{x+3}{(x-2)(x-1)(x+1)}dx = \int \frac{A}{(x-2)} + \frac{B}{(x-1)} + \frac{C}{x+1} = A(x-1)(x+1) + B(x-2)(x+1) + C(x-2)(x-1)$

$x=2$

$8-2\cdot4-2+2=0$

$x=1$

$1^3-2\cdot1^2-1+2=0$

$x=-1$

$-1^3-2(-1)^2-(-1)+2=0$

$x=2$

$2+3 = A(2-1)(2+1) + B(0) + C(0)$

$5 = A(1)(3)$

$A = \frac{5}{3}$

$= \int \frac{5/3}{(x-2)} + \frac{-2}{(x-1)} + \frac{1/3}{(x+1)}dx$

$= \frac{5}{3}\ln|x-2| - 2\ln|x-1| + \frac{1}{3}\ln|x+1| + K$

$x=1$

$1+3 = A(0) + B(1-2)(1+1) + C(0)$

$4 = B(-1)(2)$

$B = -2$

$x=-1$

$-1+3 = A(0) + B(0) + C(-1-2)(-1-1)$

$2 = C(-3)(-2)$

$C = \frac{2}{6} = \frac{1}{3}$

animativa

f- $\int \frac{x^2+1}{(x-2)^3}\,dx = \int \frac{x^2+1}{(x-2)^3}\,dx = \int \frac{(u+2)^2+1}{u^3}\,du = \int \frac{u^2+4u+4+1}{u^3}\,du$

$u = x-2 \Rightarrow x = u+2$

$du = dx$

$= \int \frac{u^2}{u^3} + \frac{4u}{u^3} + \frac{5}{u^3}\,du$

$= \int \frac{1}{u} + \frac{4}{u^2} + \frac{5}{u^3}\,du$

$= \ln|u| - \frac{4}{u} - \frac{5}{2u^2} + K$

$= \ln|x-2| - \frac{4}{x-2} - \frac{5}{2(x-2)^2} + K$

g- $\int \frac{x^5+3}{x^3-4x}\,dx = \int x^2+4 + \frac{-16x+3}{x^3-4x}\,dx = \int x^2+4\,dx + \int \frac{-16x+3}{x^3-4x}\,dx$

$x^5+0x^4+0x^3+0x^2+0x+3 \;|\; x^3-4x$

$-x^5 \quad -4x^3 \qquad\qquad x^2+4$

$4x^3+0x^2+0x+3$

$-4x^3 \quad -16x$

$-16x+3$

x^3-4x

$x(x^2-4)$

$x.(x-2)(x+2)$

$= \int x^2+4\,dx + \int \frac{-16x+3}{x(x-2)(x+2)}$

$= \int x^2+4\,dx + \int \frac{A}{x} + \frac{B}{(x-2)} + \frac{C}{(x+2)}$

$= \int x^2+4\,dx + \int A(x-2)(x+2) + B(x)(x+2) + C(x)(x-2)$

$x=0$

$-16.0+3 = A(-2)(2) + B(0) + C(0)$

$3 = A(-4)$

$A = -\frac{3}{4}$

$x=-2$

$-16.-2+3 = A(0)+B(0)+C(-2)(-4)$

$35 = C(8)$

$C = \frac{35}{8}$

$x=2$

$-16.2+3 = A(0)+B(2)(4)+C(0)$

$-29 = B(8)$

$B = -29/8$

$= \frac{x^3}{3} + 4x + \int \frac{-3/4}{x} + \frac{-29/8}{(x-2)} + \frac{35/8}{(x+2)}$

$= \frac{x^3}{3} + 4x - \frac{3}{4}\ln|x| - \frac{29}{8}\ln|x-2| + \frac{35}{8}\ln|x+2| + K$

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2-a. $\dfrac{x-3}{(x-1)^2(x+2)^2} = \dfrac{A}{(x-1)} + \dfrac{B}{(x-1)^2} + \dfrac{C}{(x+2)} + \dfrac{D}{(x+2)^2} \Rightarrow A(x-1)(x+2)^2 + B(x+2)^2 + C(x-1)^2(x+2) + D(x-1)^2$

$x=1$

$1-3 = A(0) + B(1+2)^2 + C(0) + D(0)$

$B = \dfrac{-2}{9}$

$x=-2$

$-2-3 = A(0) + B(0) + C(0) + D(-2-1)^2$

$D = \dfrac{-5}{9}$

$x=0$

$-3 = -4A + 4B + 2C + D$

$-3 = -4A + 4 \cdot \dfrac{-2}{9} + 2C + \left(\dfrac{-5}{9}\right)$

$-2A + C = \dfrac{-7}{9}$

$x=-1$

$-4 = -2A + B + 4C + 4D$

$-4 = -2A + 4C - \dfrac{2}{9} + 4\left(\dfrac{-5}{9}\right)$

$-2A + 4C = \dfrac{-14}{9}$

$-2A + 4C - (-2A + C) = \dfrac{-14}{9} - \left(\dfrac{-7}{9}\right)$

$3C = \dfrac{-7}{9}$

$C = \dfrac{-7}{27}$

$-2A - \dfrac{7}{27} = \dfrac{-7}{9}$

$A = \dfrac{7}{27}$

$= \displaystyle\int \dfrac{7/27}{(x-1)} + \dfrac{-2/9}{(x-1)^2} - \dfrac{7/27}{x+2} - \dfrac{5/9}{(x+2)^2}\,dx$

$= \dfrac{7}{27}\ln|x-1| + \dfrac{2}{9(x-1)} - \dfrac{7}{27}\ln|x+2| + \dfrac{5}{9(x+2)} + K$

3. $\displaystyle\int \dfrac{x^5+x+1}{x^3-8} = \dfrac{x^3}{3} + \int \dfrac{8x^2+x+1}{x^3-8}\,dx \Rightarrow \dfrac{8x^2+x+1}{x^3-8} = \dfrac{A}{x-2} + \dfrac{Bx+C}{x^2+2x+4}$

$x^5+x+1 \,|\, x^3-8$

x^2

$8x^2+x+1$

x^3-2^3

$A(x^2+2x+4) + Bx+C(x-2)$

$x=2 \qquad x=0$

$35 = 12A \qquad 1 = 4A - 2C$

$A = \dfrac{35}{12} \qquad C = \dfrac{16}{3}$

$x=1 \Rightarrow 10 = 7A - B - C \Rightarrow B = 61/12$

$= \dfrac{x^3}{3} + \displaystyle\int \dfrac{35/12}{x-2} + \dfrac{\frac{61x}{12} + 16/3}{x^2+2x+4}$ $\qquad u = x+1$

$= \dfrac{x^3}{3} + \dfrac{35}{12}\ln|x-2| + \dfrac{61}{24}\ln|x^2+2x+4| + \dfrac{5\sqrt{3}}{12}\operatorname{arctg}\dfrac{x+1}{\sqrt{3}} + K$

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